

When standard validation checks pass but the construct is still invalid: a worked failure case in cross-corpus multimodal mental-health machine learning

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Abstract

Cross-corpus multimodal models in mental-health machine learning fuse corpora that share no participants, and are typically validated by a battery of four internal checks: label shuffling, noise injection, a fusion ladder, and expected calibration error. We present a fully documented case in which all four passed on a classifier that measured nothing about the construct in its title.

A bipolar-state classifier built from four public corpora reached 82.6% macro-F1 and satisfied every check. A provenance-preserving reanalysis identifies three independent mechanisms, each sufficient on its own to produce that result. Rows were paired across modalities by class label and nothing else, so permuting one modality within classes left performance unchanged, at 0.828 against 0.828. The physiological feature vector could not have been measured for any participant in its source dataset, because its two halves exist for disjoint groups of people; the resulting missingness pattern alone predicts the derived label at 0.542 macro-F1 against a chance floor of 0.333. The physiological labels came from a hand-written mapping under which the largest class is undefined time between the conditions of a laboratory protocol. Two further hypotheses were tested and rejected, and are reported as negative results.

Rebuilt correctly, the physiological corpus supports a modest result: 0.523 macro-F1 for three-class diagnosis under subject-grouped evaluation, against a chance floor of 0.333.

The four checks share a blind spot. Each perturbs an assembled feature matrix, and none inspects how that matrix was assembled. We propose a seven-item reporting checklist that closes this gap, and report where its diagnostics returned null in our own analysis.

1 Introduction

In mental-health machine learning, the use of cross-corpus multimodal models is proliferating. These models fuse physiological signals (such as actigraphy from wrist-worn devices or other wearable sensors) with text (social-media posts or clinical notes) and are increasingly applied to detect or predict mental-health states, including the phases of bipolar disorder (euthymia, depression, and mania). In many cases the different modalities are drawn from separate public corpora that share no participants.

Authors typically rely on a small set of internal validation checks: label shuffling (randomly re-assigning the class labels and re-training), noise injection, a “fusion ladder” (comparing text-only, physiology-only, and fused performance), and calibration metrics such as Expected Calibration Error (ECE), which quantifies how well a model’s predicted probabilities match observed frequencies. These checks can all pass even when the assembled feature matrix contains no genuine cross-modal information about the claimed clinical construct.

We present a fully documented failure case. A multimodal classifier for bipolar states (euthymia / depression / mania) reached 82.6% macro-F1 (the unweighted average of the per-class F1 scores) and passed all four conventional checks. A provenance-preserving reanalysis—i.e., a re-run of the pipeline that retains full information about how every row and label was created—shows that three independent mechanisms each fully explain the result: (1) vacuous within-class pairing, (2) source-file identity laundered into the labels via vertical concatenation and mean imputation, and (3) a hand-mapped physiolog-

ical label scheme that does not survive inspection. None of these mechanisms requires real bipolar-state information.

This paper makes three contributions:

1. A complete, reproducible postmortem of a multimodal mental-health classifier that appeared successful by standard metrics.
2. Identification of three concrete mechanisms that defeat conventional validation checks.
3. A practical seven-item reporting checklist that would have detected each of the failures.

The work analysed here is our own, from a draft manuscript that was never submitted. We note this because it determines what the paper can offer: access to the pipeline’s full history, the labelling decisions and the discarded intermediate steps, rather than inference from a published description.

2 Background: validation conventions in multimodal mental-health ML

The four checks that the original pipeline passed are now routine in multimodal mental-health machine learning. Label shuffling randomly re-assigns the class labels and re-trains the model; a sharp drop in performance is taken as evidence that the model was using real label information. Noise injection replaces or corrupts one modality; if performance falls to the level of the remaining modality, that modality is judged to have been contributing. The fusion ladder compares text-only, physiology-only and fused performance, and incremental gains are interpreted as evidence of complementary cross-modal information. Expected calibration error quantifies how well predicted probabilities match observed frequencies, and a low value is taken as evidence that the model is well calibrated.

These checks appear frequently in the multimodal mental-health literature and are often treated as sufficient evidence that a model is using genuine signal. Recent surveys note the rapid growth of architectures fusing physiological, textual and behavioural modalities while highlighting persistent weaknesses in evaluation practice, including fragmented benchmarks, variable data-splitting conventions, and limited attention to calibration and transportability [Al Sahili et al., 2024]. Reviews of machine learning for precision psychiatry have similarly emphasised the promise of multimodal approaches while calling for more rigorous validation [Chen et al., 2022].

Each check answers a precise question. The difficulty is that each is read as answering a broader one, as set out in Table 1.

Table 1: What each conventional check establishes, and what it is commonly read as establishing.

Check	What it establishes	What it is read as establishing
Label permutation	The features carry information about the label	The model has learned genuine structure
Fusion ladder	Performance differs across modality subsets	Fusion contributes complementary information
Noise injection	Information content matters, not dimensionality	The modality carries signal
Calibration (ECE)	Confidence tracks accuracy on this test set	Uncertainty estimates are deployable

The right-hand column is the inferential step this paper examines. The gap is not a flaw in any individual check, each of which is valid for its own question. It arises because all four take the assembled feature matrix as given, and none has access to the facts that would reveal how that matrix was assembled.

Three literatures bear on this directly. The first concerns models that learn properties of data provenance rather than of the clinical target. Zech et al. [2018] trained pneumonia classifiers on chest radiographs from three hospital systems and found internal performance consistently exceeded external performance; the same networks could identify the originating hospital system in over 99.9% of images. A model able to recognise where data came from can use that information in place of the pathology it is supposed to detect. The mechanism we report in Section 5 is a tabular, cross-corpus instance of the same failure, in which the identifying signal is not an imaging artefact but the pattern of which columns a row’s source file happens to supply.

The second concerns leakage and evaluation design. Kapoor and Narayanan [2023] survey seventeen fields, identify leakage in 294 papers, and present a taxonomy of eight leakage types ranging from textbook errors to open research problems, together with model info sheets intended to catch them before publication. Within mental health specifically, Lee et al. [2023] show that incorrect cross-validation inflates the apparent diagnostic performance of deep-learning systems after data augmentation, and Moreno-Alcayde et al. [2024] show in affective computing that how data are partitioned largely determines what performance is observed.

The third concerns the choice of cross-validation unit. Saeb et al. [2017] demonstrate, on both real and simulated data, that record-wise cross-validation massively overestimates accuracy relative to subject-wise cross-validation when multiple records come from each participant, and that nearly half of the studies they retrieved using wearables or smartphones for clinical prediction used the record-wise form. A companion commentary [Little et al., 2017] argues that subject-wise validation is not a universal substitute and that the appropriate unit depends on the intended use case. We report both forms throughout for this reason, rather than treating either as correct by default.

What these literatures share is a concern with how data were assembled and partitioned rather than with how a fitted model behaves. Conventional internal validation remains necessary, but without explicit scrutiny of data assembly, label provenance and pairing procedure, high scores on the four checks can coexist with models that measure source identity, missingness patterns or other artefacts rather than the claimed clinical construct. That is the gap the present case study is designed to illustrate, and the reason our contribution is a set of things to compute about a dataset rather than a further check to run on a model.

3 The original pipeline

3.1 Goal of the original project

The original pipeline aimed to classify bipolar states by fusing physiological signals with mental-health text. Public corpora were used because clinical multimodal data with shared participants is scarce and largely restricted to the groups that collect it. The resulting classifier was validated with the internal checks conventional in the field.

3.2 The four corpora

Four public corpora were used. No participant appears in more than one, and no corpus supplies more than one modality.

WESAD provides physiological signals from chest- and wrist-worn devices, segmented into one-second windows, for 15 subjects (S2–S17, excluding S12) [Schmidt et al., 2018]. Participants followed a laboratory protocol of baseline rest, amusement, stress induced by the Trier Social Stress Test, and two meditation periods. Condition codes 0–7 are supplied per sample, where code 0 denotes undefined or transitional time.

OBF-Psychiatric provides minute-level motor activity for 162 participants across five groups: ADHD (45), a mixed clinical sample (40), depression (23), schizophrenia (22) and controls (32) [Garcia-Ceja et al., 2025].

Kaggle bipolar Reddit provides 1,692 posts with no diagnostic labels [Kaggle Reddit bipolar dataset]. **Multi-class depression Twitter** provides 11,237 posts after mapping, with categories assigned by the corpus authors [Multi-class Depression Dataset].

The target variable, bipolar state, is supplied by none of them. Labels were therefore derived separately for each corpus, as summarised in Table 2.

Table 2: Label provenance for each corpus.

Corpus	Label assigned by	Method
WESAD	Corpus authors, remapped	Hand mapping of condition codes
OBF-Psychiatric	The pipeline	Gaussian mixture over autoencoder latents
Kaggle Reddit	The pipeline	k -means on BERT embeddings, named by VADER
Twitter	Corpus authors, remapped	Hand mapping of categories

Three of the four label sets were therefore assigned by the pipeline itself or by hand mapping, not by clinicians.

OBF-Psychiatric contains no bipolar diagnostic group. Its only bipolar information is a per-participant flag in two of its metadata tables, which the pipeline did not use.

3.3 Fusion procedure

Physiological rows were assembled first. WESAD windows were class-balanced by resampling each class to the size of the smallest, using a function whose default samples with replacement; 29.3% of the resulting pool therefore consists of duplicated rows. Within each class, equal numbers of WESAD and OBF rows were then drawn without replacement and concatenated in draw order, producing a 34-dimensional physiological vector of 26 WESAD features and 8 OBF features.

Text and physiological rows were fused by the same procedure. For each class, the smaller of the two available counts determined the number of fused rows; rows were drawn without replacement and concatenated in draw order.

A fused row therefore consists of one text post and one unrelated physiological recording that share a class label. Nothing else links them: they come from different corpora, different people and different recording contexts.

Class sizes were equalised to the smallest available, 47, set by the smallest OBF mixture component; yielding 141 rows of 802 dimensions, 47 per class.

3.4 Model and evaluation protocol

All models were random forests of 200 trees with balanced class weights at a fixed seed, evaluated on a single stratified 80/20 split giving 29 test rows. Three models were reported: text-only, fitted on the full text corpus; physiological-only; and fusion. The text-only baseline was therefore fitted on a different set of rows from the fusion model, an asymmetry we return to in Section 5.

Probability calibration used isotonic regression fitted on the training rows of a pre-fitted forest. Expected calibration error was computed on the 29 test rows and reported at five bins, where the analysis code’s default is ten.

Four checks were reported as passed: label shuffling, which cost 5.3 points against a threshold of $\geq 5\%$; the fusion ladder, incremental against a threshold of $\geq 3\%$; noise injection, which degraded performance to the text-only baseline; and expected calibration error of 0.048 against a threshold of < 0.15 . The three numeric thresholds were not pre-registered and were set after the corresponding results were available.

4 Standard validation results

All three reported figures reproduce exactly. Re-scoring the original saved models on their saved test sets gives macro-F1 of 0.756 for text, 0.823 for physiological signals and 0.826 for fusion; refitting from the stored arrays on the same split returns 0.826.

These figures are not in error, and Section 5 does not revise them. The reanalysis that follows reproduces every one of them and then shows what they do not establish.

All four validation checks passed on their own terms (Table 3).

Table 3: The four conventional checks as applied to the original pipeline.

Check	Result	Threshold	Outcome
Label shuffling	5.3-point drop	$\geq 5\%$ drop	Pass
Fusion ladder	Incremental across subsets	$\geq 3\%$ gain	Pass
Noise injection	Degraded to text-only baseline	Qualitative	Pass
Calibration (ECE)	0.048	< 0.15	Pass

Each was correctly implemented. Nothing in this section is a technical error, and the reanalysis that follows does not identify one.

Two details of the protocol are recorded here because they bear on interpretation. The three numeric thresholds were not pre-registered; each was set once the corresponding result was available. The bin count used for the reported calibration error was changed from the analysis code’s default of ten to five after inspection.

How can four independent checks pass on a model that measures nothing about the construct in its title? Because they are not independent in the way that matters. Each perturbs an assembled feature matrix and observes how a fitted model responds, and none examines how that matrix was assembled including which rows were paired with which, where the labels came from, or whether the feature vector could have been measured for anyone at all.

5 Diagnostic reanalysis

The original pipeline reported 82.6% macro-F1 and passed four conventional checks. A provenance-preserving reanalysis accounts for this score through three independent mechanisms, none of which requires genuine information about bipolar state. Sections 5.3 and 5.5 report two further confounds that compound these mechanisms without being independent of them. Two hypotheses were also tested and ruled out; we report them as negative results to strengthen the record.

5.1 Vacuous pairing: no cross-modal information existed

Rows from the text and physiological corpora were paired at random within each class. Nothing was shared across modalities except the class label, so performance remained unchanged when one modality was permuted within each class. Figure 1 gives the comparison under repeated five-fold cross-validation.

The conventional across-class permutation costs 7.6 points under cross-validation, and 5.3 points on the single split used in the original report, because it destroys the label information each modality carries on its own. The within-class permutation re-pairs the two modalities while holding each row’s class label fixed, so it preserves that label-marginal information and removes only the association between modalities. If the original pairing was driven by the label alone, the joint distribution is unchanged. Performance is identical to three decimal places (Figure 1).

The same pattern holds at every stage of the correction waterfall (Figure 2). No alignment claim was supportable at any point.

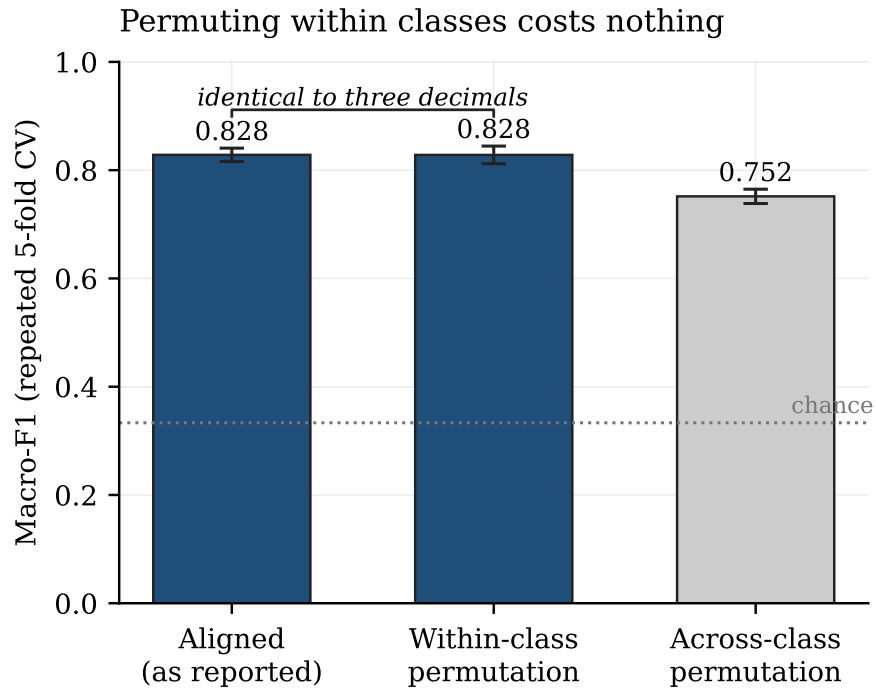


Figure 1: Macro-F1 under aligned pairing, within-class permutation and across-class permutation, under repeated five-fold cross-validation. The aligned and within-class conditions are identical to three decimal places. Error bars are standard deviations across repeats; the dotted line marks the chance floor.

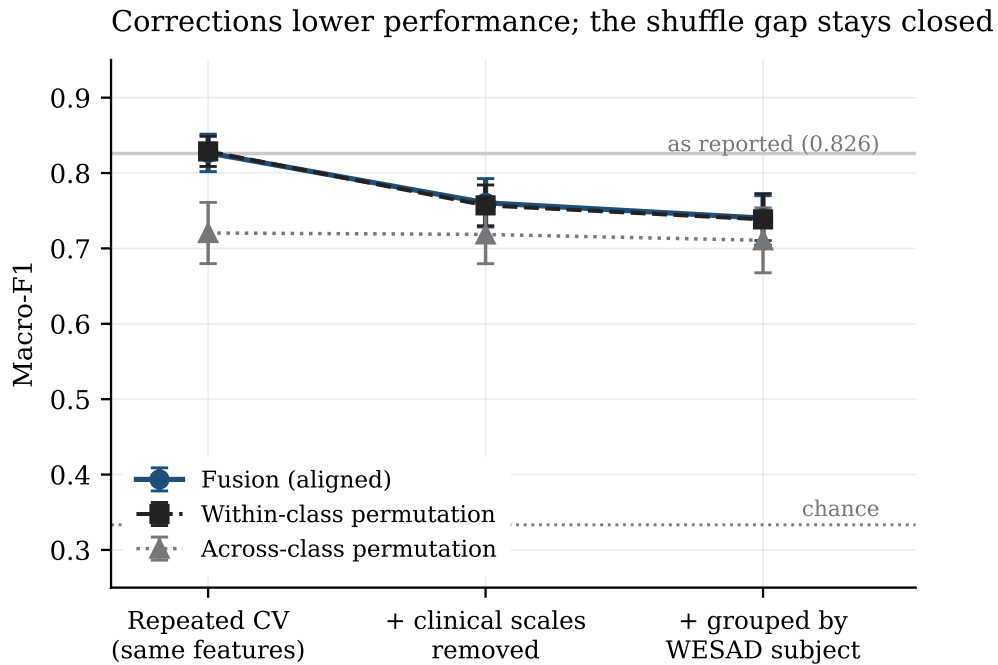


Figure 2: Fusion performance and its within-class permutation across cumulative corrections. The two upper series are superimposed at every stage and are not distinguishable in the plot. Error bars are standard deviations across ten independent redraws of the pairing.

The standard shuffle test permutes across all classes and therefore cannot distinguish a dataset in which the modalities are genuinely related from one in which the only link between them is the shared label. That is the blind spot which allows vacuous pairing to pass unnoticed.

5.2 Source-file identity laundered into the labels (OBF-Psychiatric)

The OBF distribution contains a derived feature table (`features.csv`) covering only the 77 people in the control, depression and schizophrenia groups, together with per-group metadata tables carrying clinical rating scales for the remaining 85 people in the ADHD and clinical groups. The two sets of participants are disjoint.

Instead of joining these files on the participant key, the pipeline concatenated them vertically and filled the resulting missing values with column means. No participant ever possessed a complete eight-feature vector: every row was half real measurement and half column mean, and the source file could be identified by which half was constant (Figure 3).

No participant has both halves of the feature vector

	<i>actigraphy</i>				<i>clinical rating scales</i>			
<code>features.csv</code> (n=77)	1.00	1.00	1.00	1.00	—	—	—	—
<code>adhd-info.csv</code> (n=45)	—	—	—	—	0.93	0.91	0.98	0.96
<code>clinical-info.csv</code> (n=40)	—	—	—	—	0.95	0.92	0.92	0.90
<code>control-info.csv</code> (n=32)	—	—	—	—	—	—	—	—
<code>depression-info.csv</code> (n=23)	—	—	—	—	—	—	—	—
<code>schizophrenia-info.csv</code> (n=22)	—	—	—	—	—	—	—	—
	mean	sd	pctZeros	median	madrs	hads_d	asrs	mdq_pos

Figure 3: Feature coverage in the concatenated OBF table, by source file. Cells give the fraction of rows in which each feature is present; a dash indicates the feature is absent from that file entirely. The actigraphy and clinical-scale blocks are supplied by disjoint sets of participants, so no participant possesses a complete eight-feature vector.

The consequences are measurable. The missingness pattern alone, with all values discarded, predicts the Gaussian mixture label at 0.542 ± 0.005 macro-F1 against a chance floor of 0.333. Forty-six of the 47 rows in the mixture component that became “Mania” came from the two clinical-only files; that component’s size of 47 is what capped the fused dataset at 141 rows. Within the fused data, the “Mania” class contains only two unique actigraphy vectors across its 47 rows, and the “Euthymia” class only one unique clinical-score vector across its 47 rows.

The Adjusted Rand Index between the mixture components and the diagnoses already present in the dataset is 0.059, which is essentially chance. OBF-Psychiatric contains no bipolar patients in this diagnosis field; an explicit bipolar indicator exists for the 85 ADHD and clinical participants, but the pipeline did not use it.

High measured performance was therefore obtained in the absence of any genuine bipolar-state signal. The mixture model recovered the source-file structure that concatenation and mean imputation had encoded, and the random forest then recovered the mixture labels from the same features that had

been used to create them. The labels are recoverable from their own inputs at 0.938 ± 0.010 . That is the circularity.

5.3 Source confounding in the text corpora

The same confound appears on the text side in milder form. A classifier predicting which corpus a post came from, using the BERT embeddings alone, reaches 0.936 macro-F1: Reddit posts and tweets are trivially separable. The two corpora also carry different label distributions, so corpus identity is partially informative about the target.

The Kaggle labels are additionally circular by construction. Assigned by k -means over the embeddings subsequently used as model inputs, they are recovered exactly, at 1.000, by nearest-centroid assignment. A model trained on these features and labels is in part re-learning the clustering procedure.

Predicting each corpus’s majority label without using the text at all gives 0.251, so corpus identity does not by itself explain the reported text performance. It is a contributing confound rather than a complete account.

5.4 Physiological label mapping did not survive inspection (WESAD)

The mapping of WESAD condition codes onto bipolar states was written by hand and corresponds to no coherent clinical or physiological grouping. Table 4 shows what it assigns.

Table 4: Hand-mapped bipolar states against actual WESAD conditions, in one-second windows (as-run 11-subject copy).

Assigned state	Actual WESAD conditions (windows)
Euthymia	transient/undefined 29,139; amusement 4,089
Depression	baseline 12,896; excluded code 6: 583
Mania	meditation 8,640; stress 7,361

Three observations follow. First, the largest class consists mostly of undefined time between protocol conditions. Second, the “manic” class pools the two conditions furthest apart in arousal, stress and meditation. Third, evaluated on the real WESAD condition labels, leave-one-subject-out cross-validation, in which each fold holds out all data from one person, yields 0.424 against a chance floor of 0.333, while a random-window split on the same data reaches 0.962. Adjacent one-second windows from one participant are near-identical, so a random split places near-copies on both sides of it. The split, not the signal, produced most of the apparent performance.

Four of the 15 subjects, S6 to S9, were missing their processed `.pkl` files in the copy available to the original pipeline. The loader reported each skip to standard output and continued, so every original WESAD figure rests on 11 of 15 subjects, or 73% of the available data. The missing files were recoverable from the dataset archive, and both the as-run and the complete analyses are published with this paper.

5.5 The fusion ladder compared unlike baselines

Under repeated cross-validation, fusion does not exceed the stronger single modality at any stage of the correction waterfall. At the first stage, physiological features alone give 0.879 ± 0.029 against fusion’s 0.827 ± 0.025 ; after the clinical scales are removed, 0.810 ± 0.035 against 0.761 ± 0.032 . The text modality is a net cost.

The original fusion ladder did not surface this because its text-only baseline was fitted on the full text corpus rather than on the 141 fused rows, so the two figures were not comparable. Where a ladder compares models fitted on different sets of rows, its increments do not measure what they appear to measure.

5.6 Hypotheses tested and rejected

Both of the following mechanisms are present in the code but did not reach the final fused dataset. We report them because the diagnostics proposed in Section 6 are offered as tools, and a reader needs to know how they behave when the thing they look for is absent.

The first hypothesis was duplicate-row leakage from resampling with replacement. The function `sklearn.utils.resample` defaults to sampling with replacement, and 29.3% of the balanced physiological pool consists of duplicated rows. However, 0.0% of test rows have an exact duplicate in the training set, for the text block, the physiological block and the WESAD sub-block alike. The duplication is real, but the fusion step samples without replacement from an already-balanced pool, so it does not reach the fused data.

The second hypothesis was that fully-imputed rows drove a class. Seventy-eight OBF rows had all eight features missing and were filled entirely with column means, which should produce a constant vector repeating within one class. The prediction held only in part: all 78 did fall into a single component, but it was not the component under suspicion, and the suspected class contains 47 distinct vectors. The hypothesis was rejected.

Pursuing it is nevertheless what revealed the mechanism in Section 5.2. The clustering was driven not by rows missing everything, but by rows missing complementary halves, a structure that a test for repeated whole vectors cannot detect because the measured half differs from row to row. The diagnostic that failed indicated where to look next.

5.7 What the data actually support

Showing that a dataset does not support a claim is incomplete without establishing what it does support. OBF-Psychiatric was therefore rebuilt from its raw activity series, with person-day features computed for all 162 participants, joined on the participant key, and filtered to complete days of 1,440 minutes with non-zero activity and no duplicate timestamps. Of 2,206 raw person-days, 335 were incomplete, 87 recorded no activity and 6 contained duplicated minutes, leaving 1,785. Every value in the published derived table appears among the computed values for each of five participants checked, confirming the feature definitions.

Table 5: Corrected OBF benchmark under subject-grouped and random splits.

Task	Macro-F1	Chance
5-class diagnosis, subject-grouped	0.314 ± 0.008	0.200
5-class diagnosis, random person-day split	0.400 ± 0.004	0.200
3-class (control/depression/schizophrenia), subject-grouped	0.523 ± 0.016	0.333

The results are modest and above chance. The 0.086 gap between the subject-grouped and random splits on identical data is attributable entirely to evaluation design.

Two genuine same-person pairings were then tested as positive controls for the within-class permutation. Actigraphy with clinical scales for 71 participants gave 0.732 aligned against 0.726 ± 0.027 permuted, a gap of +0.006. Co-registered WESAD chest and wrist sensors gave 0.422 ± 0.019 aligned against 0.429 ± 0.017 permuted, a gap of +0.007. Both are null. In the first case the clinical scales alone give 0.733 against fusion’s 0.732, so actigraphy contributes nothing for the permutation to disturb. In the second, one-second summary features of chest and wrist both track a slowly varying arousal state, so the precise instant of pairing carries little information.

The conclusion we draw is narrower than the one we sought. The within-class permutation is diagnostic of a claim rather than of a mechanism: a null result establishes that no alignment claim is supportable, which is what it establishes for the original pipeline. Demonstrating that it responds to genuine cross-modal dependence requires data in which that dependence is known and adjustable.

A final caution concerns the interaction of the two diagnostics. Permuting one modality across all rows under subject-grouped cross-validation assigns a held-out row its second modality from a training participant, which reintroduces exactly the leakage the grouping was imposed to remove. The permuted score then rises to 0.592 ± 0.025 against an aligned 0.422. A permutation that improves performance is a leakage signature, not evidence of alignment, and the permutation must be confined to cells defined jointly by participant and class.

6 A practical reporting checklist

All four conventional checks passed, yet three independent mechanisms produced an 82.6% macro-F1 that measured nothing about bipolar state. Those checks took the assembled feature matrix as given and never inspected how its rows or labels were created. The following seven items close that gap. Each is inexpensive to compute and maps onto a failure mode documented in Section 5.

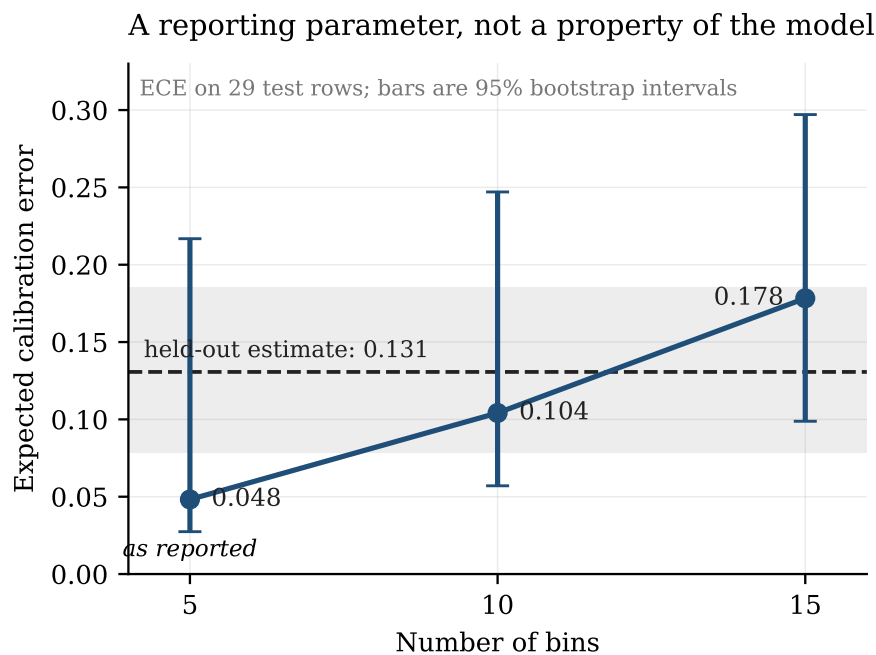


Figure 4: Expected calibration error on the 29 test rows as a function of bin count, with 95% bootstrap intervals. The reported figure used five bins. The dashed line and shaded band give the estimate obtained when the calibrator is fitted by internal cross-validation and evaluated on held-out data.

1. **Shared-participant statement.** State explicitly whether any individual appears in more than one modality. In the original pipeline the four corpora shared no participants, so the fusion was pure label-based pairing.
2. **Pairing and join method.** Report the exact procedure used to combine rows from different modalities. Here, for each of the three labels, equal numbers of text and physiological rows were drawn without replacement and concatenated in draw order. A fused row consisted of one text post and one unrelated physiological recording that happened to share a label, and nothing else linked them. This produced a balanced set of 141 rows, 47 per class.
3. **Label provenance for every modality.** For each label set, state who or what assigned the labels, and whether the features later used as model input were involved in creating them.

- WESAD: condition codes from the corpus authors, remapped by hand.

- OBF-Psychiatric: assigned by the pipeline, by Gaussian mixture over autoencoder latents.
- Kaggle bipolar Reddit: assigned by the pipeline, by k -means over BERT embeddings, named by VADER sentiment.
- Multi-class depression Twitter: corpus labels, remapped by hand.

Three of the four label sets were therefore assigned by the pipeline or by hand mapping, not by clinicians. Both sets of pipeline-assigned labels were derived from features later fed back into the classifier. The hand-mapped WESAD labels were never checked against the source codebook.

4. **Within-class permutation test.** In addition to the conventional across-class shuffle, report a within-class permutation of one modality. Here the across-class shuffle cost 7.6 points under cross-validation ($0.828 \rightarrow 0.752$), while the within-class permutation left performance unchanged at 0.828 ± 0.016 , demonstrating that the pairing carried no cross-modal information. Under grouped evaluation, confine the permutation to cells defined by participant and class.
5. **Subject-grouped evaluation.** Report performance under subject-grouped or leave-one-subject-out cross-validation alongside any random-split figure. On the corrected OBF data the subject-grouped five-class macro-F1 was 0.314 ± 0.008 against 0.400 ± 0.004 for a random person-day split. On WESAD, leave-one-subject-out on the real condition labels gave 0.424 against 0.962 for a random-window split. Random splits substantially inflated apparent performance in both cases.
6. **Missingness and uniqueness audit.** Report the proportion of rows containing imputed values, whether any class is dominated by a small number of unique feature vectors, and the predictive power of the missingness mask alone. The OBF feature table was built by vertical concatenation followed by mean imputation, and no participant possessed a complete eight-feature vector. The missingness pattern alone predicted the mixture label at 0.542 ± 0.005 macro-F1 against a chance floor of 0.333. Within the fused data the “Mania” class contained two unique actigraphy vectors across 47 rows, and “Euthymia” one unique clinical-score vector across 47 rows.
7. **Calibration reporting standard.** Report expected calibration error at more than one bin count, with interval estimates, and where possible fit the calibrator by internal cross-validation rather than by a single post-hoc fit on a small test set. The reported ECE of 0.048 was obtained with five bins on 29 test rows, and its 95% bootstrap interval extends to 0.217. The same predictions give 0.104 at ten bins and 0.178 at fifteen (Figure 4). Fitted by internal cross-validation and evaluated held-out, ECE is 0.131 [0.078, 0.185]. The headline figure was a reporting artefact.

One further item concerns joins rather than models. A merge validated by counting null values in its own key will always appear to succeed, because a key column is never null by construction. Validate instead on a column supplied only by the table being joined in.

7 Limitations

A single case. We analyse one pipeline, built by us from four corpora. The three mechanisms are demonstrated, not shown to be prevalent, and how often cross-corpus fusion in this literature rests on them remains an empirical question this paper does not answer. Nothing here implies that published work using these checks shares these flaws; our claim concerns what the checks are able to detect, not who has been caught by them.

The diagnostics are validated only negatively. The within-class permutation test returned null on both genuine same-person pairings available to us, so this paper demonstrates no sensitivity for it on real data. What a null result does license is a specific negative conclusion, that no alignment claim is supportable, and that is the finding for the original pipeline. Showing that the test responds to genuine cross-modal dependence requires data in which that dependence is known and adjustable, which these corpora do not provide.

A reconstruction, not the original execution. PyTorch was unseeded in the original run and library defaults have changed since, so intermediate quantities may differ marginally from those originally produced. All headline figures nevertheless reproduce to three decimal places.

The corrected benchmark is small. It rests on 162 participants and 1,785 person-days of actigraphy from a single corpus. It establishes a comparator rather than a clinical finding, and we make no claim about its generalisation.

Bipolar subtypes are not analysed. An affective-type code in one metadata table is not defined in the corpus documentation, and we have not relied on it. The explicit bipolar indicator used in its place covers 85 participants.

The checklist is not prospectively validated. Its seven items are derived from a single case and argued on general grounds; none has been applied to independent work before the fact.

8 Conclusion

A multimodal classifier for bipolar states reported 82.6% macro-F1 and passed four standard validation checks. Its physiological feature vector could not have been measured for any participant in its source dataset, because the two halves of that vector exist for disjoint groups of people. Its rows were paired across modalities by class label and by nothing else. Its largest physiological class consisted mostly of undefined time between the conditions of a laboratory protocol.

None of the four checks could have revealed any of this. Each one perturbs an assembled feature matrix and observes how a fitted model responds, while all three facts above concern how that matrix came to exist. The checks were not applied carelessly, and they did not fail on their own terms. They were simply answering a different question from the one their results were taken to answer.

What follows from this is less daunting than it might appear. The corrective is not a better model, a stricter threshold, or a longer battery of the same kind of check. It is a short list of things worth computing about a dataset before fitting anything to it: whether its rows describe people, whether its labels were derived from the features those labels are attached to, whether its splits separate participants, and whether its pattern of missing values carries the signal. Each of these takes minutes to compute. In this case, any one of them would have been decisive.

We report the corrected analysis alongside the failure because the two belong together. Rebuilt properly, the physiological corpus supports a real result: three-class diagnosis at 0.523 macro-F1 under subject-grouped evaluation, against a chance floor of 0.333, on 162 participants and a single modality. That is a smaller claim than the one we began with, and it is one the data can carry. The distance between it and 82.6% is the subject of this paper, and closing that distance cost nothing but the willingness to look at how the data had been assembled.

Data and code availability

All analysis code, both sets of results, and the derived person-day feature table are available at <https://github.com/divyashie/multimodal-BD-Detection>. Every figure reported here is

written to a machine-readable results file in that repository and can be regenerated without re-running the analysis. The four source corpora are not redistributed; WESAD and OBF-Psychiatric are available from their authors under their own terms.

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